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## SYSTEM AND METHOD FOR CONSUMING ACIDIC FLUIDS IN WELLS

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### Abstract

A method of consuming an acidic fluid in a wellbore comprises injecting a multilayer metallic ball into the wellbore, wherein the multilayer metallic ball comprises a core layer, a metallic layer encapsulating the core layer, and a ball density greater than the acidic fluid; contacting the multilayer metallic ball with the acidic fluid, thereby reacting at least a portion of the metallic layer with the acidic fluid at a first reaction rate and at least a portion of the core layer with the acidic fluid at a second reaction rate; and converting at least a portion of the acidic fluid to a spent acidic fluid.

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### Background/Summary

## FIELD

[0001] The present disclosure relates to methods and systems of consuming acidic fluids, and particularly to methods and systems of consuming acidic fluids in a wellbore.

## BACKGROUND

[0002] Acid treatment (also referred to herein as acidizing) is commonly used to improve well productivity in oil and gas industry. Acidizing typically involves pumping an acidic fluid such as hydrogen chloride (HCl) or hydrogen fluoride (HF) into a wellbore and/or geologic formation to clean out debris or formation damage restricting flow in the wellbore or geologic formation. Other applications of acidizing (commonly referred to as fracture acidizing) may be used to dissolve formation rock in the geologic formation to create new flow paths to the wellbore or enlarge existing ones. Upon completion of acid treatment, acids in the acidic fluid are consumed and a spent fluid is produced. The spent fluid is typically benign.

## SUMMARY

[0003] However, if acid treatment is incomplete, for example, as a result of tight formation, acids in the acidic fluid may react with the casing or tubular materials, cause corrosion and material losses, and lead to compromised structural integrity, casing leakage, abnormal annular pressure, and other serious concerns. In this case, the acidic fluid should be immediately removed. The unreacted acidic fluid may be circulated out the well into an acid tank above the surface and subsequently neutralized in the acid tank. However, the removal and neutralization processes pose great safety threats to field operators. Additionally, immediate removal of the unreacted acidic fluid may not be practically feasible until a removal device, such as a coiled tubing, is available and can be deployed into the well. Until then, the acidic fluid remains in the well and continues reacting with the casing and tubular materials.

[0004] Accordingly, a need exists to develop methods and systems that are capable of consuming unreacted acids directly in a wellbore to minimize damages to the wellbore and safety threats to the field operators.

[0005] The present disclosure is directed to methods of and systems for consuming acids in an acidic fluid in a wellbore. For example, an unreacted acidic fluid may be converted to a spent acidic fluid in the wellbore before being circulated out the well, and therefore, the need for subsequent acid neutralization in an acid tank is diminished and therefore the safety threats to the field operators are minimized. Further, once the acidic fluid has been converted to the spent acidic fluid, the spent acidic fluid is less reactive than the acidic fluid and thus less likely to react with the casing or tubular material and damage the wellbore.

[0006] Embodiments of the present disclosure address these needs by utilizing at least one multilayer metallic ball to consume an acidic fluid in a wellbore and, in particular, at least one multilayer metallic ball that is capable of outcompeting and/or impeding the reaction between the acid fluid and metal components in the wellbore, such as a tubing or a casing.

[0007] For example, the casing, the tubing, or combinations thereof may react with the acidic fluid at a third reaction rate ( $R_{\text{sub.3}}$ ). In embodiments, the multilayer metallic balls may comprise a core layer and a metallic layer encapsulating the core layer. The metallic layer may be configured to react with the acidic fluid at a first reaction rate ( $R_{\text{sub.metal}}$ ) greater than or equal to 5 times and less than or equal to 200 times of the third reaction rate ( $R_{\text{sub.3}}$ ), and the core layer may be configured to react with the acidic fluid at a second reaction rate ( $R_{\text{sub.core}}$ ) greater than or equal to 100 times and less than or equal to 5000 times of the third reaction rate ( $R_{\text{sub.3}}$ ). The multilayer metallic balls may further comprise a ball density greater than the acidic fluid. Therefore, upon injecting the multilayer metallic ball into the acidic fluid in the wellbore, the multilayer metallic ball will begin reacting with the acidic fluid and simultaneously sinking downwards in the wellbore. While sinking downwards, the metallic ball will continue reacting with the acidic fluid.

[0008] In accordance with one embodiment of the present disclosure, a method of consuming an

acidic fluid in a wellbore may comprise injecting at least one multilayer metallic ball into the wellbore, wherein the at least one multilayer metallic ball comprises a core layer, a metallic layer encapsulating the core layer, and a ball density greater than the acidic fluid; contacting the at least one multilayer metallic ball with the acidic fluid, thereby reacting at least a portion of the metallic layer with the acidic fluid at a first reaction rate ( $R_{\text{sub.metal}}$ ) and exposing at least a portion of the core layer; and contacting the core layer with the acidic fluid, thereby reacting at least a portion of the core layer with the acidic fluid at a second reaction rate ( $R_{\text{sub.core}}$ ) and consuming at least a portion of the core layer, wherein reacting at least a portion of the metallic layer or the core layer with the acidic fluid converts at least a portion of the acidic fluid to a spent acidic fluid and forms a mixture of the acidic fluid and the spent acidic fluid.

[0009] In accordance with another embodiment of the present disclosure, a system for consuming an acidic fluid in a wellbore may comprise the wellbore; the acidic fluid in the wellbore; and at least one multilayer metallic ball comprising a core layer, a metallic layer encapsulating the core layer, and a ball density greater than the acidic fluid, wherein the metallic layer is configured to react with the acidic fluid at a first reaction rate ( $R_{\text{sub.metal}}$ ) and thereby convert a portion of the acidic fluid to a spent acidic fluid, the acidic fluid and the spent acidic fluid forming a mixture, and the core layer is configured to react with the acidic fluid at a second reaction rate ( $R_{\text{sub.core}}$ ) and thereby convert another portion of the acidic fluid to additional spent acidic fluid.

[0010] It is to be understood that both the foregoing general description and the following detailed description describe various embodiments and are intended to provide an overview or framework for understanding the nature and character of the claimed subject matter. The accompanying drawings are included to provide a further understanding of the various embodiments, and are incorporated into and constitute a part of this specification. The drawings illustrate the various embodiments described herein, and together with the description serve to explain the principles and operations of the claimed subject matter.

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## Description

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0011] The following detailed description of specific embodiments of the present disclosure can be best understood when read in conjunction with the following drawings, where like structure is indicated with like reference numerals and in which:

[0012] FIG. 1A illustrates a system for consuming an acidic fluid in a wellbore;

[0013] FIG. 1B illustrates a cross-sectional view of the wellbore;

[0014] FIG. 1C illustrates a multilayer metallic ball for consuming the acidic fluid in the wellbore;

[0015] FIG. 2 illustrates reacting the multilayer metallic ball with the acidic fluid in the wellbore; and

[0016] FIG. 3 illustrates a method of consuming the acidic fluid in the wellbore; specifically,

[0017] FIG. 3A illustrates the acidic fluid is being injected into the wellbore;

[0018] FIG. 3B illustrates at least one multilayer metallic ball is introduced into the wellbore contacting the acidic fluid, reacting with the acidic fluid, converting the acidic fluid to a spent acidic fluid; and forming a mixture of the acidic fluid and the spent acidic fluid;

[0019] FIG. 3C illustrates a displacement is injected into the wellbore upon determining completions of consuming the acidic fluid; and

[0020] FIG. 3D illustrates the mixture of the acidic fluid and the spent acidic fluid is being displaced by the displacement and flows toward to a surface of the wellbore.

### DETAILED DESCRIPTION

[0021] As used herein, “acid” includes, but is not limited to, a Brønsted-Lowry acid, a proton donor, or an electron acceptor. Examples of acids may comprise inorganic acids, such as sulfuric,

nitric, hydrochloric, hydrofluoric, and phosphoric acids, as well as organic acids, such as carboxylic acids, sulfonic acids, and phenol groups.

[0022] As used herein, “consume” or “consuming” an acidic fluid means a chemical reaction or a physical process in which acids in the acidic fluid are being reduced or removed.

[0023] As used throughout this disclosure, the terms “downward” and “upward” may refer to a position within a wellbore relative to the surface, with upward indicating direction or position closer to the surface and downward referring to direction or position farther away from the surface.

[0024] As described in the present disclosure, a “geologic formation” may refer to a body of rock that is sufficiently distinctive and continuous from the surrounding rock bodies that the body of the rock may be mapped as a distinct entity. A geologic formation is, therefore, sufficiently homogenous to form a single identifiable unit containing similar properties throughout the geologic formation, including, but not limited to, porosity and permeability.

[0025] As used herein, “impeding” a reaction means that the reaction is hindered, obstructed, inhibited, blocked, or delayed by another competing chemical reaction or by a physical process that consumes or removes a reactant of the reaction or that produces or adds a product of the reaction such that the reaction is disfavored.

[0026] As used herein, “outcompeting” a reaction means that the reaction is surpassed, defeated, reduced, outdone, or displaced by another competing reaction that is more effective or aggressive. For example, if the reaction and the competing reaction have a common reactant (e.g. acids), the occurrence of the reaction may become less probable if the number density (e.g. concentration or surface area) of the common reactant is reduced by the faster, more effective, more aggressive competing reaction.

[0027] As used throughout this disclosure, “wellbore,” may refer to a drilled hole or borehole extending from the surface of the Earth down to the geologic formation, including the openhole or uncased portion. The wellbore may form a pathway capable of permitting fluids to traverse between the surface and the geologic formation. The wellbore may include at least a portion of a fluid conduit that links the interior of the wellbore to the surface. The fluid conduit connecting the interior of the wellbore to the surface may be capable of permitting regulated fluid flow from the interior of the wellbore to the surface and may permit access between equipment on the surface and the interior of the wellbore.

[0028] Reference will now be made in detail to embodiments of systems for consuming an acidic fluid in a wellbore.

[0029] Referring initially to FIG. 1A, according to embodiments, a system **100** for consuming an acidic fluid **300** in a wellbore **110** may comprise the wellbore **110**; the acidic fluid **300** in the wellbore **110**; and at least one multilayer metallic ball **200**. In some embodiments, the wellbore **110** may also comprise a casing **120**, a tubing **140**, or combinations thereof. In some embodiments, the system **100** may further comprise a wellhead **150** connected to the wellbore **110**, such as by the casing **120**, the tubing **140**, or both. As shown in FIG. 1B, and in one or more embodiments, the wellbore **110** may further comprise a wellbore annulus **130** defined by an inner diameter of the wellbore **110** or the casing **120** and an outer diameter of the tubing **140**. In some embodiments, the wellbore annulus **130** may be defined by the inner diameter of the wellbore **110** and an outer diameter of the casing **120**. In other embodiments, the wellbore **110** may comprise a plurality of casings **120**, and the wellbore annulus **130** may be defined by an inner diameter of a casing and an outer diameter of another casing.

[0030] Referring now to FIG. 1C, the at least one multilayer metallic ball **200** may comprise a core layer **240** and a metallic layer **220** encapsulating the core layer **240**. The at least one multilayer metallic ball **200** may also comprise a ball density greater than the acidic fluid. The metallic layer **220** may be configured to react with the acidic fluid **300** at a first reaction rate (R.sub.metal) and thereby convert a portion of the acidic fluid **300** to a spent acidic fluid **320**. The acidic fluid **300** and the spent acidic fluid **320** may together form a mixture. The core layer **240** may also be

configured to react with the acidic fluid **300** at a second reaction rate (R.sub.core) and thereby convert another portion of the acidic fluid **300** to additional spent acidic fluid **320**.

[0031] In embodiments, the metallic layer **220** and the core layer **240** may be made from materials that may effectively outcompete and/or impede the reaction between the acidic fluid **300** and the at least a portion of the casing **120**, the tubing **140**, or combinations thereof; that is, a rate of the reaction between the acidic fluid **300** and the at least one multilayer metallic ball **200** is greater than a rate of the reaction between the acidic fluid **300** and the wellbore **110**, such as the casing **120**, the tubing **140**, or combinations thereof.

[0032] For example, and in embodiments, the casing **120**, the tubing **140**, or combinations thereof may react with the acidic fluid **300** at a third reaction rate (R.sub.3). The first reaction rate (R.sub.metal) may be relatively greater than the third reaction rate (R.sub.3), such that the metal layer may be consumed within 18 hours, such as within 15 hours, within 12 hours, or within 6 hours. The second reaction rate (R.sub.core) may also be relatively greater than the third reaction rate (R.sub.3), such that the core layer may be consumed within 24 hours, such as within 18 hours, within 12 hours, within 6 hours, or within 3 hours.

[0033] Without being limited by theory, the first and second reaction rates being greater than the third reaction rates may enable the depletion or reduction of acids in the acidic fluid **300** and thus slow or stop the reaction between the acidic fluid and the casing **120**, the tubing **140**, or combinations thereof, minimizing the chances of wellbore corrosion, material loss, or loss of wellbore integrity.

[0034] Further, a reaction rate may be concentration-dependent. That is, once the metallic layer **220** begins to react with a portion of the acidic fluid **300**, the concentration of acids in the acidic fluid **300** begins to reduce, and therefore the first rate (R.sub.metal) of reaction between the acidic fluid **300** and the metallic layer **220** may begin to reduce. Therefore, in order to maintain the overall reactivity between the at least one multilayer metallic ball **200** and the acidic fluid **300**, the second reaction rate (R.sub.core) between the acidic fluid **300** and the core layer **240** may be greater than the first reaction rate (R.sub.metal) between the acidic fluid **300** and the metallic layer **220** and further greater than the third reaction rate (R.sub.3) between the acidic fluid **300** and the casing **120**, the tubing **140**, or combinations thereof.

[0035] Other variables such as the composition and surface area may also affect the reaction rate. In one or more embodiments, the metallic layer **220** may comprise a metallic material. The metallic material of the metallic layer **220** may comprise alkaline earth metals, aluminum, tin, zinc, cerium, neodymium, iron, nickel, lead, or combinations thereof. In further embodiments, the metallic layer may further comprise a non-metallic material, such as a blend of metallic and non-metallic materials. The non-metallic material of the metallic layer **220** may comprise oxides, carbonates, polymers, or combinations thereof. The oxides may be selected from one or more of silica, alumina, iron oxides, cerium oxides, neodymium oxides, titanium oxides, andradite, alkaline metal oxides, copper oxides, and manganese oxide. The carbonates may be selected from one or more of calcium carbonates, magnesium carbonates, and iron carbonates. The polymers may be selected from one or more of polysaccharides, polyacrylic acids, polylactic acids, and poly(vinylalcohol)s. For example, and in embodiments, the metallic layer **220** may comprise a blend of alkaline earth metals and aluminum. In further embodiments, the metallic layer **220** may comprise aluminum and a blend of calcium carbonates, polysaccharides, polyacrylic acids, and combinations thereof. In other further embodiments, the metallic layer **220** may comprise alkaline earth metals and a blend of calcium carbonates, polysaccharides, polyacrylic acids, and combinations thereof.

[0036] Where the metallic layer **220** comprises a blend of the metallic material and the non-metallic material, the metallic material may comprise from 5 wt. % to 99.9 wt. % of the total weight of the metallic layer, such as 5 wt. % to 10 wt. %, from 10 wt. % to 30 wt. %, from 30 wt. % to 50 wt. %, from 50 wt. % to 70 wt. %, from 70 wt. % to 99.9 wt. %, or any combination of the previous ranges or smaller range therein, such as from 5 wt. % to 25 wt. %, 5 wt. % to 20 wt. %, 5

wt. % to 15 wt. %, from 10 wt. % to 25 wt. %, from 10 wt. % to 20 wt. %, from 10 wt. % to 15 wt. %, from 15 wt. % to 25 wt. %, from 15 wt. % to 20 wt. %, or from 20 wt. % to 25 wt. %.

[0037] Now reference to the composition of the core layer **240**, according to embodiments, the core layer **240** may comprise a metallic material, a non-metallic material, or combinations thereof.

[0038] In one or more embodiments, the metallic material of the core layer **240** may comprise alkaline earth metals, aluminum, tin, zinc, cerium, neodymium, iron, nickel, lead, or combinations thereof.

[0039] In one or more embodiments, the non-metallic material of the core layer **240** may comprise oxides, carbonates, polymers, or combinations thereof. The oxides may be selected from one or more of silica, alumina, iron oxides, cerium oxides, neodymium oxides, titanium oxides, andradite, alkaline metal oxides, copper oxides, and manganese oxide. The carbonates may be selected from one or more of calcium carbonates, magnesium carbonates, and iron carbonates. The polymers may be selected from one or more of polysaccharides, polyacrylic acids, polylactic acids, and poly(vinylalcohol)s.

[0040] For example, in further embodiments, the core layer **240** may comprise polymers comprising from 0.1 wt. % to 95 wt. % of the total weight of the core layer, such as from 0.1 wt. % to 15 wt. %, from 15 wt. % to 30 wt. %, from 30 wt. % to 40 wt. %, from 40 wt. % to 50 wt. %, from 50 wt. % to 70 wt. %, from 70 wt. % to 95 wt. %, or any combination of the previous ranges or smaller range therein, such as from 10 wt. % to 35 wt. %, from 10 wt. % to 30 wt. %, from 10 wt. % to 25 wt. %, from 10 wt. % to 20 wt. %, from 10 wt. % to 15 wt. %, from 15 wt. % to 35 wt. %, from 15 wt. % to 30 wt. %, from 15 wt. % to 25 wt. %, from 15 wt. % to 20 wt. %, from 20 wt. % to 35 wt. %, from 20 wt. % to 30 wt. %, from 20 wt. % to 25 wt. %, from 25 wt. % to 35 wt. %, from 25 wt. % to 30 wt. %, or from 30 wt. % to 35 wt. %.

[0041] In one or more embodiments, the core layer **240** may comprise a combination of the metallic material and the non-metallic material.

[0042] For example, and in further embodiments, the metallic material may comprise 5 wt. % to 99.9 wt. % of the total weight of the core layer **240**, such as from 5 wt. % to 10 wt. %, from 10 wt. % to 30 wt. %, from 30 wt. % to 50 wt. %, from 50 wt. % to 70 wt. %, from 70 wt. % to 99.9 wt. %, or any combination of the previous ranges or smaller range therein, such as from 5 wt. % to 20 wt. %, from 5 wt. % to 15 wt. %, from 5 wt. % to 10 wt. %, from 5 wt. % to 8 wt. %, from 8 wt. % to 20 wt. %, from 8 wt. % to 15 wt. %, from 8 wt. % to 10 wt. %, from 10 wt. % to 20 wt. %, from 10 wt. % to 15 wt. %, or from 15 wt. % to 20 wt. %. For example, and in other embodiments, the non-metallic material may comprise from 0.1 wt. % to 95 wt. % of the total weight of the core layer **240**, such as from 0.1 wt. % to 15 wt. %, from 15 wt. % to 30 wt. %, 30 wt. % to 40 wt. %, from 40 wt. % to 50 wt. %, from 50 wt. % to 70 wt. %, from 70 wt. % to 95 wt. %, or any combination of the previous ranges or smaller range therein, from 10 wt. % to 35 wt. %, from 10 wt. % to 30 wt. %, from 10 wt. % to 25 wt. %, from 10 wt. % to 20 wt. %, from 10 wt. % to 15 wt. %, from 15 wt. % to 35 wt. %, from 15 wt. % to 30 wt. %, from 15 wt. % to 25 wt. %, from 15 wt. % to 20 wt. %, from 20 wt. % to 35 wt. %, from 20 wt. % to 30 wt. %, from 20 wt. % to 25 wt. %, from 25 wt. % to 35 wt. %, from 25 wt. % to 30 wt. %, or from 30 wt. % to 35 wt. %.

[0043] In addition to the compositions stated hereinabove, it is also contemplated that the metallic layer **220** and the core layer **240** may individually further comprise at least one additive such as an acid inhibitor. The acid inhibitor may hinder, inhibit, block, or slow down the reaction between the acid fluid **300** and the metal components (e.g. the casing **120** or the tubing **140**) in the wellbore **110**, and therefore protect these metal components from the corrosive acidic fluid **300**.

[0044] Regarding the dimension of the at least one multilayer metallic ball **200**, according to embodiments, the at least one multilayer metallic ball **200** may comprise an initial radius of from 0.05 cm to 5 cm before the at least one multilayer metallic ball **200** is injected into the wellbore, such as from 0.05 cm to 1 cm, from 0.1 cm to 0.5 cm, from 0.5 cm to 1 cm, from 1 cm to 1.5 cm, from 1.5 cm to 2 cm, from 2 cm to 2.5 cm, from 2.5 cm to 3.5 cm, from 3 cm to 3.5 cm, from 3.5

cm to 4 cm, from 4 cm to 4.5 cm, from 4.5 cm to 5 cm, or any combination of the previous ranges or smaller range therein, such as from 2 cm to 5 cm, from 2 cm to 4.5 cm, from 2 cm to 4 cm, from 2 cm to 3.5 cm, from 2 cm to 3 cm, from 2 cm to 2.5 cm, from 2.5 cm to 5 cm, from 2.5 cm to 4.5 cm, from 2.5 cm to 4 cm, from 2.5 cm to 3.5 cm, from 2.5 cm to 3 cm, from 3 cm to 5 cm, from 3 cm to 4.5 cm, from 3 cm to 4 cm, from 3 cm to 3.5 cm, from 3.5 cm to 5 cm, from 3.5 cm to 4.5 cm, from 3.5 cm to 4 cm, from 4 cm to 5 cm, from 4 cm to 4.5 cm, or from 4.5 cm to 5 cm.

[0045] Further, as stated hereinabove, surface areas of individual layers may affect the reaction rate. Without being bound by any particular theory, increasing the surface area-to-volume ratio of materials may exposing more of the materials to the surrounding environment and therefore increasing reactivity or accelerating chemical reactions of these materials.

[0046] Therefore, according to embodiments, the at least one multilayer metallic ball **200** may individually comprise an initial radius of from 5 nm to 500  $\mu\text{m}$  before the at least one multilayer metallic ball **200** is injected into the wellbore, such as from 5 nm to 200 nm, from 200 nm to 500  $\mu\text{m}$ , from 500 nm to 1  $\mu\text{m}$ , from 1  $\mu\text{m}$  to 50  $\mu\text{m}$ , from 50  $\mu\text{m}$  to 100  $\mu\text{m}$ , from 100  $\mu\text{m}$  to 150  $\mu\text{m}$ , from 150  $\mu\text{m}$  to 200  $\mu\text{m}$ , from 200  $\mu\text{m}$  to 250  $\mu\text{m}$ , from 250  $\mu\text{m}$  to 300  $\mu\text{m}$ , from 300  $\mu\text{m}$  to 350  $\mu\text{m}$ , from 350  $\mu\text{m}$  to 400  $\mu\text{m}$ , from 400  $\mu\text{m}$  to 450  $\mu\text{m}$ , from 450  $\mu\text{m}$  to 500  $\mu\text{m}$ , or any combination of the previous ranges or smaller range therein, such as from 100  $\mu\text{m}$  to 250  $\mu\text{m}$ , from 100  $\mu\text{m}$  to 200  $\mu\text{m}$ , from 100  $\mu\text{m}$  to 175  $\mu\text{m}$ , from 100  $\mu\text{m}$  to 150  $\mu\text{m}$ , from 150  $\mu\text{m}$  to 250  $\mu\text{m}$ , from 150  $\mu\text{m}$  to 200  $\mu\text{m}$ , from 150  $\mu\text{m}$  to 175  $\mu\text{m}$ , from 175  $\mu\text{m}$  to 250  $\mu\text{m}$ , from 175  $\mu\text{m}$  to 200  $\mu\text{m}$ , or from 200  $\mu\text{m}$  to 250  $\mu\text{m}$ .

[0047] Varying the material structures may also increase surface areas of the metallic layer **220** and/or the core layer **240**. Therefore, in embodiments, the metallic layer **220** and the core layer **240** may individually comprise various material structures such as crystalline material, porous material, amorphous material, or agglomerates. For example, the metallic layer **220** and the core layer **240** may be individually formed by compacting a plurality of nanoparticles comprising compositions described hereinabove, and upon contacting an acidic fluid and initiating the acid degradation, the compacts or agglomerates of nanoparticles would break into individual particles or groups of particles increasing the total surface area and thus acid reactivity.

[0048] For example, and in embodiments, the at least one multilayer metallic ball **200** may comprise an initial radius of from 0.05 cm to 5 cm before the at least one multilayer metallic ball **200** is injected into the wellbore, and the metallic layer **220** and the core layer **240** of the at least one multilayer metallic ball **200** may be individually formed by compacting nanoparticles of metallic materials, non-metallic materials, or both. In further embodiments, the nanoparticles of metallic materials, non-metallic materials, or both may comprise a diameter of from 5 nm to 200 nm, such as from 5 nm to 20 nm, from 20 nm to 40 nm, from 40 nm to 60 nm, from 60 nm to 80 nm, from 80 nm to 100 nm, from 100 nm to 200 nm, or any combination of the previous ranges or smaller range therein, such as from 5 nm to 175 nm, from 5 nm to 150 nm, from 5 nm to 125 nm, from 5 nm to 100 nm, from 5 nm to 75 nm, from 5 nm to 50 nm, from 5 nm to 25 nm, from 25 nm to 175 nm, from 25 nm to 150 nm, from 25 nm to 125 nm, from 25 nm to 100 nm, from 25 nm to 75 nm, from 25 nm to 50 nm, from 50 nm to 175 nm, from 50 nm to 150 nm, from 50 nm to 125 nm, from 50 nm to 100 nm, from 50 nm to 75 nm, from 75 nm to 175 nm, from 75 nm to 150 nm, from 75 nm to 125 nm, from 75 nm to 100 nm, from 100 nm to 175 nm, from 100 nm to 150 nm, from 100 nm to 125 nm, from 125 nm to 175 nm, from 125 nm to 150 nm, or from 150 nm to 175 nm.

[0049] Regarding the density of the at least one multilayer metallic ball **200**, the at least one multilayer metallic ball **200** may initially comprise a ball density greater than the fluid density so that the at least one multilayer metallic ball **200** may sink to the bottom of the wellbore **110** or a horizontal section of the wellbore, begin to react with the acidic fluid while descending down the wellbore **110** and continue to react with the acidic fluid at the bottom of the wellbore **110** or in the horizontal section of the wellbore **110**.

[0050] According to embodiments, the at least one multilayer metallic ball **200** may initially comprise the ball density (d.sub.Ball) of from 1.2 g/cm.<sup>3</sup> to 8.9 g/cm.<sup>3</sup>; and the acidic fluid **300** may comprise the fluid density (d.sub.fluid) of from 1.0 g/cm.<sup>3</sup> to 1.9 g/cm.<sup>3</sup> before the at least one multilayer metallic ball **200** contacts the acidic fluid and start reacting with the acidic fluid. For example, the initial ball density may be from 1.2 g/cm.<sup>3</sup> to 7.5 g/cm.<sup>3</sup>, from 1.2 g/cm.<sup>3</sup> to 5.0 g/cm.<sup>3</sup>, from 1.2 to 2.5 g/cm.<sup>3</sup>, from 2.5 g/cm.<sup>3</sup> to 8.9 g/cm.<sup>3</sup>, from 2.5 g/cm.<sup>3</sup> to 7.5 g/cm.<sup>3</sup>, from 2.5 g/cm.<sup>3</sup> to 5.0 g/cm.<sup>3</sup>, from 5.0 g/cm.<sup>3</sup> to 8.9 g/cm.<sup>3</sup>, from 5.0 g/cm.<sup>3</sup> to 7.5 g/cm.<sup>3</sup>, or from 7.5 g/cm.<sup>3</sup> to 8.9 g/cm.<sup>3</sup>. For example, the initial fluid density may be from 1.0 g/cm.<sup>3</sup> to 1.7 g/cm.<sup>3</sup>, from 1.0 g/cm.<sup>3</sup> to 1.5 g/cm.<sup>3</sup>, from 1.0 g/cm.<sup>3</sup> to 1.3 g/cm.<sup>3</sup>, from 1.3 g/cm.<sup>3</sup> to 1.7 g/cm.<sup>3</sup>, from 1.3 g/cm.<sup>3</sup> to 1.5 g/cm.<sup>3</sup>, or from 1.5 g/cm.<sup>3</sup> to 1.7 g/cm.<sup>3</sup>. [0051] The density of the at least one multilayer metallic ball **200** may be altered by varying the composition of the metallic layer **220** and/or the composition of the core layer **240** such that the at least one multilayer metallic ball **200** may sink, suspend, or float in the fluid at different stages of degradation.

[0052] For example, and in embodiments, a core density (d.sub.core) of the core layer **240** may be greater than the fluid density (d.sub.fluid) such that, after a portion of the metallic layer **220** has been consumed, the at least one multilayer metallic ball **200** or the core layer **240** may continue descending down the wellbore **110**, remaining sunk at the bottom of the wellbore **110**, and therefore reaching the bottom of the wellbore **110** to consume the acidic fluid **300** there. According to embodiments, the core layer **240** may comprise a core density (d.sub.core) from 0.92 g/cm.<sup>3</sup> to 1.5 g/cm.<sup>3</sup>, from 0.92 g/cm.<sup>3</sup> to 1.1 g/cm.<sup>3</sup>, or from 1.1 g/cm.<sup>3</sup> to 1.5 g/cm.<sup>3</sup>; and the core density (d.sub.core) is less than or equal to the fluid density d.sub.fluid such that the core layer **240** suspends in the acidic fluid **300**, the spent acidic fluid **320**, the displacement fluid **340**, or combinations thereof. For example, and in one or more embodiments, the suspending or floating core layer **240** may comprise polymers comprising from 70 wt. % to 95 wt. % of the total weight of the core layer **240**, such as from 70 wt. % to 75 wt. %, from 75 wt. % to 80 wt. %, from 80 wt. % to 85 wt. %, from 85 wt. % to 90 wt. %, from 90 wt. % to 95 wt. %, or any combination of the previous ranges or smaller range therein, such as from 70 wt. % to 90 wt. %, from 70 wt. % to 85 wt. %, from 70 wt. % to 80 wt. %, from 75 wt. % to 95 wt. %, from 75 wt. % to 85 wt. %, from 80 wt. % to 95 wt. %, from 80 wt. % to 90 wt. %, or from 85 wt. % to 95 wt. %.

[0053] For another example, the core density (d.sub.core) may be less than or equal to the fluid density (d.sub.fluid) such that, after a portion of the metallic layer has been consumed, the at least one multilayer metallic ball **200** or the core layer **240** may suspend or float in the mixture of the acidic fluid **300** and the spent acidic fluid **320**. Subsequently, the at least one partially-consumed multilayer metallic ball **200** or the core layer **240** may be circulated along with the mixture fluid or the displacement fluid to the surface of the wellbore **110** to be removed or recollected. Also, during the removal process, the at least one multilayer metallic ball **200** or core layer **240** may continue reacting with residual acids in the mixture fluid. According to embodiments, the core layer **240** may comprise a core density (d.sub.core) from 1.5 g/cm.<sup>3</sup> to 7.2 g/cm.<sup>3</sup>, from 1.5 g/cm.<sup>3</sup> to 3.5 g/cm.<sup>3</sup>, or from 3.5 g/cm.<sup>3</sup> to 7.2 g/cm.<sup>3</sup>; and the core density (d.sub.core) is greater than the fluid density d.sub.fluid such that the core layer **240** sinks in the acidic fluid **300**. In this instance, and in one or more embodiments, the sinking core layer **240** may comprise metallic materials, oxides, carbonates, or combinations thereof comprising greater than 30 wt. % of the total weight of the core layer, such as greater than 50 wt. %, greater than 70 wt. %, greater than 90 wt. %, greater than 95 wt. %, from 30 wt. % to 50 wt. %, from 50 wt. % to 70 wt. %, from 70 wt. % to 90 wt. %, from 90 wt. % to 95 wt. %, from 95 wt. % to 99.9 wt. %, or any combination of the previous ranges or smaller range therein, such as from 85 wt. % to 99.9 wt. %, from 85 wt. % to 95 wt. %, from 85 wt. % to 90 wt. %, or from 90 wt. % to 99.9 wt. %.

[0054] Regarding the density of the metallic layer **220**, in one or more embodiments, the metallic



layer **220** may comprise a metal density (d.sub.metal) from 2.0 g/cm.sup.3 to 11.3 g/cm.sup.3, such as from 2.0 g/cm.sup.3 to 9.0 g/cm.sup.3, from 2.0 g/cm.sup.3 to 6.0 g/cm.sup.3, from 2.0 g/cm.sup.3 to 4.5 g/cm.sup.3, from 4.5 g/cm.sup.3 to 11.3 g/cm.sup.3, from 4.5 g/cm.sup.3 to 9.0 g/cm.sup.3, from 4.5 g/cm.sup.3 to 6.0 g/cm.sup.3, from 6.0 g/cm.sup.3 to 11.3 g/cm.sup.3, from 6.0 g/cm.sup.3 to 9.0 g/cm.sup.3, or from 9.0 g/cm.sup.3 to 11.3 g/cm.sup.3.

[0055] The density of the metallic layer **220** may be adjusted by varying the metallic layer **220** composition by blending metallic materials with non-metallic materials. For example, and in further embodiments, the metallic material may comprise 5 wt. % to 99.9 wt. % of the total weight of the metallic layer **220**, such as from 5 wt. % to 10 wt. %, from 10 wt. % to 30 wt. %, from 30 wt. % to 50 wt. %, from 50 wt. % to 70 wt. %, from 70 wt. % to 99.9 wt. %, or any combination of the previous ranges or smaller range therein, such as from 45 wt. % to 75 wt. %, from 45 wt. % to 70 wt. %, from 45 wt. % to 65 wt. %, from 45 wt. % to 60 wt. %, from 45 wt. % to 55 wt. %, from 45 wt. % to 50 wt. %, from 50 wt. % to 75 wt. %, from 50 wt. % to 65 wt. %, from 50 wt. % to 60 wt. %, from 50 wt. % to 55 wt. %, from 55 wt. % to 75 wt. %, from 55 wt. % to 70 wt. %, from 55 wt. % to 65 wt. %, from 55 wt. % to 60 wt. %, from 60 wt. % to 75 wt. %, from 60 wt. % to 70 wt. %, from 60 wt. % to 65 wt. %, from 65 wt. % to 75 wt. %, from 65 wt. % to 70 wt. %, or from 70 wt. % to 75 wt. %. For example, and in other embodiments, the non-metallic material may comprise from 0.1 wt. % to 95 wt. % of the total weight of the metallic layer **220**, such as from 0.1 wt. % to 15 wt. %, from 15 wt. % to 30 wt. %, from 30 wt. % to 40 wt. %, from 40 wt. % to 50 wt. %, from 50 wt. % to 70 wt. %, from 70 wt. % to 95 wt. %, or any combination of the previous ranges or smaller range therein, such as from 45 wt. % to 75 wt. %, from 45 wt. % to 70 wt. %, from 45 wt. % to 65 wt. %, from 45 wt. % to 60 wt. %, from 45 wt. % to 55 wt. %, from 45 wt. % to 50 wt. %, from 50 wt. % to 75 wt. %, from 50 wt. % to 65 wt. %, from 50 wt. % to 60 wt. %, from 50 wt. % to 55 wt. %, from 55 wt. % to 75 wt. %, from 55 wt. % to 70 wt. %, from 55 wt. % to 65 wt. %, from 55 wt. % to 60 wt. %, from 60 wt. % to 75 wt. %, from 60 wt. % to 70 wt. %, from 60 wt. % to 65 wt. %, from 65 wt. % to 75 wt. %, from 65 wt. % to 70 wt. %, or from 70 wt. % to 75 wt. %.

[0056] Reference will now be made in detail to embodiments of methods for consuming acidic fluids. The methods of consuming acidic fluids may utilize any of the systems **100** and/or multilayer metallic balls **200** previously described.

[0057] Referring now to FIG. 2, contacting the at least one multilayer metallic ball **200** with the acidic fluid **300** in the wellbore **110** may initially cause a portion of the metallic layer **220** to react with the acidic fluid **300**, as shown in the left of FIG. 2. Upon reacting the portion of the metallic layer **220** with the acidic fluid **300**, the portion of the metallic layer **220** is consumed exposing a portion of the core layer **240**, and a portion of the acidic fluid is converted to a spent acidic fluid **320** and produce a mixture of the acidic fluid **300** and the spent acidic fluid **320**. As shown in the middle of FIG. 2, at this stage, the fluid mixture may contain more the acidic fluid **300** than the spent acidic fluid **320**. Once a portion of the core layer **240** is exposed, the exposed portion of the core layer **240** is in contact with the acidic fluid **300** and continues reacting with the acidic fluid **300** and converting the acidic fluid **300** to the spent acidic fluid **320**, as shown in the right of FIG. 2. As the reaction continues, the fluidic mixture may contain more the spent acidic fluid **320** than the acidic fluid **300**. Accordingly, the fluidic mixture become less reactive to the metal components (e.g. the casing **120** or the tubing **140**) in the wellbore **110** and less hazardous to the field operator when the fluidic mixture is removed or flew back to the surface of the wellbore **110**.

[0058] Now referring to FIG. 3, illustrated are methods of consuming an acidic fluid **300** in the wellbore **110**. According to embodiments, a method of consuming an acidic fluid **300** in a wellbore **110** (FIG. 3A) may comprise injecting at least one multilayer metallic ball **200** disclosed hereinabove into the wellbore **110** disclosed hereinabove (FIG. 3B), contacting the multilayer metallic ball **200** with the acidic fluid **300** (FIG. 3B), thereby reacting at least a portion of the metallic layer **220** with the acidic fluid **300** at a first reaction rate (R.sub.metal) and exposing at least a portion of the core layer **240**, and contacting the core layer **240** with the acidic fluid **300**,

thereby reacting at least a portion of the core layer **240** with the acidic fluid **300** at a second reaction rate (R.sub.core) and consuming at least a portion of the core layer **240**.

[0059] The methods disclosed hereinabove may further comprise removing the mixture of the acidic fluid **300** and the spent acidic fluid **320**. For example, and as shown in FIG. 3C, the methods disclosed hereinabove may further comprise injecting a displacement fluid **340** into the wellbore **110**. In some embodiments, injecting the displacement fluid **340** into the wellbore **110** may enable displacing at least a portion of the mixture of the acidic fluid and the spent acidic fluid, as shown in FIG. 3D. In some embodiments, the method may further comprise injecting additional displacement fluid **340** until the mixture of the acidic fluid **300** and the spent acidic fluid **320** is removed from the wellbore **110** through the tubing **140** or the wellbore annulus **130** defined by, for example, an inner diameter of the wellbore and an outer diameter of the tubing, as shown in FIGS. 1A and 1B.

[0060] According to embodiments, the methods disclosed herein may further comprise determining the extent or completeness of acid degradation, such as by assessing an average pH of the mixture of the acidic fluid **300** and the spent acidic fluid **320**. For example, the process of consuming an acidic fluid **300** may be considered as substantially completed upon determining an average pH of the mixture of the acidic fluid **300** and the spent acidic fluid **320** is greater than or equal to 5.

[0061] In one or more embodiments, upon determining the process of consuming an acidic fluid **300** is not substantially completed or the average pH of the mixture of the acidic fluid **300** and the spent acidic fluid **320** is less than 5, the method may further comprise injecting at least one additional multilayer metallic ball **200** into the wellbore **110**.

[0062] Upon determining the process of consuming an acidic fluid is completed or the average pH of the mixture of the acidic fluid **300** and the spent acidic fluid **320** is greater than or equal to 5, the displacement or removal of the mixture of the acidic fluid and the spent fluid may commence, such as by injecting the displacement fluid **340** into the wellbore **110** as described previously. The injecting of the displacement fluid **340** into the wellbore **110** may thereby displacing at least a portion of the mixture of the acidic fluid **300** and the spent acidic fluid **320**. As previously stated, the method may further comprise injecting additional displacement fluid **340** until the mixture of the acidic fluid **300** and the spent acidic fluid **320** is removed from the wellbore **110** through the tubing **140** or the wellbore annulus **130**. In further embodiments, the wellbore may comprise an average pressure within the wellbore greater than a pressure at the surface of the wellbore, such that the mixture of the acidic fluid and the spent acidic fluid may be flowed back to the surface of the wellbore as a result of the pressure difference.

[0063] Having described the subject matter of the present disclosure in detail and by reference to specific embodiments thereof, it is noted that the various details disclosed herein should not be taken to imply that these details relate to elements that are essential components of the various embodiments described herein, even in cases where a particular element is illustrated in each of the drawings that accompany the present description. Further, it will be apparent that modifications and variations are possible without departing from the scope of the present disclosure, including, but not limited to, embodiments defined in the appended claims. More specifically, although some aspects of the present disclosure are identified herein as preferred or particularly advantageous, it is contemplated that the present disclosure is not necessarily limited to these aspects.

[0064] It is also noted that terms like “preferably,” “commonly,” and “typically,” when utilized herein, are not utilized to limit the scope of the claimed invention or to imply that certain features are critical, essential, or even important to the structure or function of the claimed invention. Rather, these terms are merely intended to identify particular aspects of an embodiment of the present disclosure or to emphasize alternative or additional features that may or may not be utilized in a particular embodiment of the present disclosure.

[0065] For the purposes of describing and defining the present invention it is noted that the terms “substantially” and “approximately” are utilized herein to represent the inherent degree of

uncertainty that may be attributed to any quantitative or qualitative comparison, value, measurement, or other representation.

[0066] It is noted that one or more of the following claims utilize the term “wherein” as a transitional phrase. For the purposes of defining the present invention, it is noted that this term is introduced in the claims as an open-ended transitional phrase that is used to introduce a recitation of a series of characteristics of the structure and should be interpreted in like manner as the more commonly used open-ended preamble term “comprising.” Likewise, utilization of the terms “having”, “has”, or their equivalents should likewise be interpreted in like manner as the more commonly used open-ended preamble term “comprising.”

## Claims

1. A method of consuming an acidic fluid in a wellbore, comprising: injecting at least one multilayer metallic ball into the wellbore, wherein the at least one multilayer metallic ball comprises a core layer, a metallic layer encapsulating the core layer, and a ball density greater than the acidic fluid; contacting the at least one multilayer metallic ball with the acidic fluid, thereby reacting at least a portion of the metallic layer with the acidic fluid at a first reaction rate (R.sub.metal) and exposing at least a portion of the core layer; and contacting the core layer with the acidic fluid, thereby reacting at least a portion of the core layer with the acidic fluid at a second reaction rate (R.sub.core) and consuming at least a portion of the core layer, wherein reacting at least a portion of the metallic layer or the core layer with the acidic fluid converts at least a portion of the acidic fluid to a spent acidic fluid and forms a mixture of the acidic fluid and the spent acidic fluid.
2. The method of claim 1, wherein: the wellbore further comprises a casing, a tubing, or combinations thereof; at least a portion of the casing, the tubing, or combinations thereof reacts with the acidic fluid at a third reaction rate (R.sub.3); the first reaction rate (R.sub.metal) is greater than the third reaction rate (R.sub.3); and the second reaction rate (R.sub.core) is greater than the third reaction rate (R.sub.3).
3. The method of claim 1, wherein: the multilayer metallic ball comprises a ball density (d.sub.Ball) of from 1.2 g/cm.sup.3 to 8.9 g/cm.sup.3; and the acidic fluid comprises a fluid density (d.sub.fluid) of from 1.0 g/cm.sup.3 to 1.9 g/cm.sup.3, such that the multilayer metallic ball sinks in the acidic fluid.
4. The method of claim 1, wherein the metallic layer has a metal density (d.sub.metal) from 2.0 g/cm.sup.3 to 11.3 g/cm.sup.3.
5. The method of claim 1, wherein the metallic layer comprises a metallic material comprising alkaline earth metals, aluminum, tin, zinc, cerium, neodymium, iron, nickel, lead, or combinations thereof.
6. The method of claim 5, wherein: the metallic layer further comprises a non-metallic material comprising oxides, carbonates, polymers, or combinations thereof; the oxides are selected from one or more of silica, alumina, iron oxides, cerium oxides, neodymium oxides, titanium oxides, andradite, alkaline metal oxides, copper oxides, and manganese oxide; the carbonates are selected from one or more of calcium carbonates, magnesium carbonates, and iron carbonates; and the polymers are selected from one or more of polysaccharides, polyacrylic acids, polylactic acids, and poly(vinylalcohol)s.
7. The method of claim 6, wherein the metallic layer comprises: the metallic material comprising from 5 wt. % to 99.9 wt. % of the total weight of the metallic layer; and the non-metallic material comprising from 0.1 wt. % to 95 wt. % of the total weight of the metallic layer.
8. The method of claim 1, wherein: the core layer comprises a metallic material, a non-metallic material, or combinations thereof; the metallic material comprises alkaline earth metals, aluminum, tin, zinc, cerium, neodymium, iron, nickel, lead, or combinations thereof; the non-metallic material

comprises oxides, carbonates, polymers, or combinations thereof; the oxides are selected from one or more of silica, alumina, iron oxides, cerium oxides, neodymium oxides, titanium oxides, andradite, alkaline metal oxides, copper oxides, and manganese oxide; the carbonates are selected from one or more of calcium carbonates, magnesium carbonates, and iron carbonates; and the polymers are selected from one or more of polysaccharides, polyacrylic acids, polylactic acids, and poly(vinylalcohol)s.

**9.** The method of claim 8, wherein: the core layer has a core density ( $d_{\text{sub.core}}$ ) of from 0.92 g/cm<sup>3</sup> to 1.5 g/cm<sup>3</sup>; and the core density ( $d_{\text{sub.core}}$ ) is less than or equal to the fluid density  $d_{\text{sub.fluid}}$  such that the core layer suspends in the acidic fluid.

**10.** The method of claim 9, wherein: the core layer comprises polymers; and the polymers comprise from 70 wt. % to 95 wt. % of the total weight of the core layer.

**11.** The method of claim 8, wherein: the core layer has a core density ( $d_{\text{sub.core}}$ ) greater than 1.5 g/cm<sup>3</sup> to 7.2 g/cm<sup>3</sup>; and the core density ( $d_{\text{sub.core}}$ ) is less than or equal to the fluid density  $d_{\text{sub.fluid}}$  such that the core layer suspends or floats in the acidic fluid.

**12.** The method of claim 11, wherein the core layer comprises metallic materials, oxides, carbonates, or combinations thereof; and the metallic materials, oxides, carbonates, or combinations thereof comprise greater than 30 wt. % of the total weight of the core layer.

**13.** The method of claim 1, further comprising assessing an average pH of the mixture of the acidic fluid and the spent acidic fluid.

**14.** The method of claim 13, wherein: the wellbore further comprises a wellbore annulus defined by an inner diameter of the wellbore and an outer diameter of the tubing; and the method further comprises, upon determining the average pH of the mixture is greater than or equal to 5, injecting a displacement fluid into the wellbore, thereby displacing at least a portion of the mixture of the acidic fluid and the spent acidic fluid, and injecting additional displacement fluid until the mixture of the acidic fluid and the spent acidic fluid is removed from the wellbore through the tubing or the wellbore annulus.

**15.** The method of claim 13, further comprising upon determining the average pH of the mixture of the acidic fluid and the spent acidic fluid is greater than or equal to 5, flowing back the mixture of the acidic fluid and the spent acidic fluid to a surface of the wellbore, wherein an average pressure within the wellbore is greater than the pressure at the surface.

**16.** The method of claim 13, further comprising upon determining the average pH of the mixture of the acidic fluid and the spent acidic fluid is less than 5, injecting at least one additional multilayer metallic ball into the wellbore.

**17.** The method of claim 1, wherein the at least one multilayer metallic ball comprises an initial radius of from 0.05 cm to 5 cm.

**18.** The method of claim 17, wherein the metallic layer and the core layer of the at least one multilayer metallic ball individually comprise nanoparticles comprising a diameter of from 5 nm to 200 nm.

**19.** A system for consuming an acidic fluid in a wellbore, comprising: the wellbore; the acidic fluid in the wellbore; and at least one multilayer metallic ball comprising a core layer, a metallic layer encapsulating the core layer, and a ball density greater than the acidic fluid, wherein the metallic layer is configured to react with the acidic fluid at a first reaction rate ( $R_{\text{sub.metal}}$ ) and thereby convert a portion of the acidic fluid to a spent acidic fluid, the acidic fluid and the spent acidic fluid forming a mixture, and the core layer is configured to react with the acidic fluid at a second reaction rate ( $R_{\text{sub.core}}$ ) and thereby convert another portion of the acidic fluid to additional spent acidic fluid.

**20.** The system of claim 19, wherein: the wellbore further comprises a casing, a tubing, or combinations thereof; the casing, the tubing, or combinations thereof reacts with the acidic fluid at a third reaction rate ( $R_{\text{sub.3}}$ ); the first reaction rate ( $R_{\text{sub.metal}}$ ) is greater than the third reaction

rate (R.sub.3); and the second reaction rate (R.sub.core) is greater than the third reaction rate (R.sub.3).

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